

Question Number	Answer	Additional Guidance	Mark
2a	An answer that includes the following: snails have a simple nervous system, so they are thought not to feel (much) pain (1)	Accept snails are invertebrates so they do not feel (much) pain / snails are not sentient beings / do not need a licence to use snails	Exp 1

Question Number	Answer	Additional Guidance	Mark
2b	An answer that includes the following: • correct use of r^3 (1) • correct multiplication by 2 (1) • correct answer to two significant figures (1)	Example calculation $1.5^3 / 3.375 / \frac{3^3}{2}$ 28 (mm³) Allow ecf if errors in first parts of calculation; 14 gets mp1 and 3; 230 gets mp2 and 3 Correct answer with no working gains full marks	Exp 3

Question Number	Answer	Additional Guidance	Mark
2d	An answer that includes two of the following • presence of water, not volume, is important (1) • rate of water supply does not determine time to break down the membrane (1) • (rate of membrane breakdown is) limited by another factor (1)	Accept once the membrane is wet, adding more water makes no difference Accept time the water has been there is important, not the rate of supply Accept there is the same rate of diffusion of water through the membrane regardless of water supply. Accept changes to membrane structure takes time Accept enzymes may be involved in the breaking of the membrane. Accept something other than water is rate limiting Accept membrane thickness may vary, so takes longer to break down (if thicker).	Exp 2